

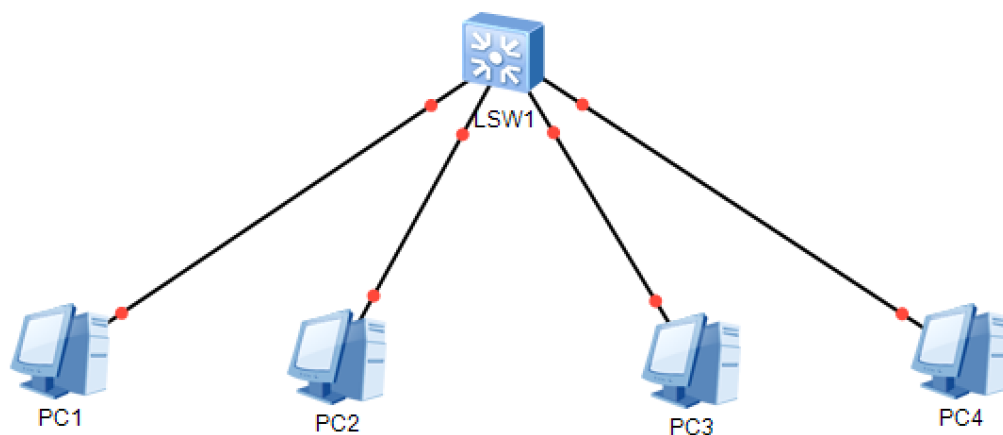
Experiment Report LAB-2

LAB-2-1: PORT-BASED VLAN CONFIGURATION

Step 4: Configure VLANs on the switch.

Answer the following questions:

Q2-1.1. Paste the screenshot of the created topology.



Q2-1.2. Draw IEEE 802.1D tagged frame format.

In IEEE 802.1D protocol, we use the standard ethernet frame format.

Destination MAC	Source MAC	EtherType	Payload	FCS
(6 bytes)	(6 bytes)	(2 bytes)	(e.g. BPDU)	(4 bytes)

Q2-1.3. Paste the screenshot of VLAN 10 information.

```
[LSW1]display vlan 10
```

U: Up;	D: Down;	TG: Tagged;	UT: Untagged;
MP: Vlan-mapping;	ST: Vlan-stacking;		
#: ProtocolTransparent-vlan;	*: Management-vlan;		

VID	Type	Ports
10	common	UT:GE0/0/9 (D) GE0/0/10 (D) GE0/0/11 (D) GE0/0/12 (D)

VID	Status	Property	MAC-LRN	Statistics	Description
10	enable	default	enable	disable	VLAN 0010

Q2-1.4. Paste the screenshot of VLAN 30 information

```
[LSW1]display vlan 30
```

U:	Up;	D:	Down;	TG:	Tagged;
MP:	Vlan-mapping;	ST:	Vlan-stacking;	UT:	Untagged;
#:	ProtocolTransparent-vlan;	*	Management-vlan;		

VID	Type	Ports			

30	common	UT:GE0/0/13 (D)	GE0/0/14 (D)	GE0/0/15 (D)	GE0/0/16 (D)

VID	Status	Property	MAC-LRN	Statistics	Description

30	enable	default	enable	disable	VLAN 0030

Q2-1.5. Suppose you are going to create a new VLAN 40, and add ports 17-20 to this VLAN in batches. Please list the configuration command.

create VLAN 40.

```
[LSW1] vlan 40
```

```
[LSW1-vlan40] quit
```

assign ports GE 0/0/17 to 0/0/20 to vlan 40 in batch

create a port group, named pgvlan40

```
[LSW1] port-group pgvlan40
```

add ports to the group

```
[LSW1-port-group-pgvlan40] group-member gigabitethernet 0/0/17 to gigabitethernet 0/0/20
```

set port link type

```
[LSW1-port-group-pgvlan40] port link-type access
```

assign ports in the group to vlan 40

```
[LSW1-port-group-pgvlan40] port default vlan 40
```

```
[LSW1-port-group-pgvlan40] quit
```

display port group information

```
[LSW1] display port-group
```

```
[LSW1] display port-group pgvlan40
```

```
[LSW1] display vlan 40
```

And finally we can get

```
[LSW1]display vlan 40
```

VID	Type	Ports
40	common	UT:GE0/0/17 (D) GE0/0/18 (D) GE0/0/19 (D) GE0/0/20 (D)

VID	Status	Property	MAC-LRN	Statistics	Description
40	enable	default	enable	disable	VLAN 0040

In the end, we display the vlan , and have.

```
[LSW1]display port vlan
```

Port	Link Type	PVID	Trunk VLAN List
GigabitEthernet0/0/1	hybrid	1	-
GigabitEthernet0/0/2	hybrid	1	-
GigabitEthernet0/0/3	hybrid	1	-
GigabitEthernet0/0/4	hybrid	1	-
GigabitEthernet0/0/5	hybrid	1	-
GigabitEthernet0/0/6	hybrid	1	-
GigabitEthernet0/0/7	hybrid	1	-
GigabitEthernet0/0/8	hybrid	1	-
GigabitEthernet0/0/9	access	10	-
GigabitEthernet0/0/10	access	10	-
GigabitEthernet0/0/11	access	10	-
GigabitEthernet0/0/12	access	10	-
GigabitEthernet0/0/13	access	30	-
GigabitEthernet0/0/14	access	30	-
GigabitEthernet0/0/15	access	30	-
GigabitEthernet0/0/16	access	30	-
GigabitEthernet0/0/17	access	40	-
GigabitEthernet0/0/18	access	40	-
GigabitEthernet0/0/19	access	40	-
GigabitEthernet0/0/20	access	40	-
GigabitEthernet0/0/21	hybrid	1	-
GigabitEthernet0/0/22	hybrid	1	-
GigabitEthernet0/0/23	hybrid	1	-
GigabitEthernet0/0/24	hybrid	1	-

Step 5: Verify the configurations.

Answer the following questions:

Q2-1.6. Can PCs in the same VLAN ping each other? Please paste the screenshot of the ping result.

Yes, they can. The results are shown as follows.

I choose the pc10-1 , whose IPv4 address is 192.168.10.11, and ping to 192.168.10.12, the other pc in the same vlan , it turns out they can.

```
PC>ping 192.168.10.12

Ping 192.168.10.12: 32 data bytes, Press Ctrl_C to break
From 192.168.10.12: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.12: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.12: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.12: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.12: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.12 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/47/47 ms
```

Q2-1.7. Can PCs in different VLANs ping each other? Please paste the screenshot of the ping result. If the ping fails, please explain the reasons.

No, they can't. I ping the pc10-1 to the pc30-1 and 30-2, the result is that they are unreachable.

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: Destination host unreachable

PC>ping 192.168.30.12

Ping 192.168.30.12: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: Destination host unreachable
```

Reason:

VLANs (Virtual LANs) are designed to segment network traffic for isolation and security. Devices in different VLANs cannot communicate with each other without a Layer 3 device (router or Layer 3 switch) to route the traffic between VLANs.

LAB-2-2: MAC ADDRESS-BASED VLAN

CONFIGURATION

Step 4: Assign MAC Addresses to VLANs.

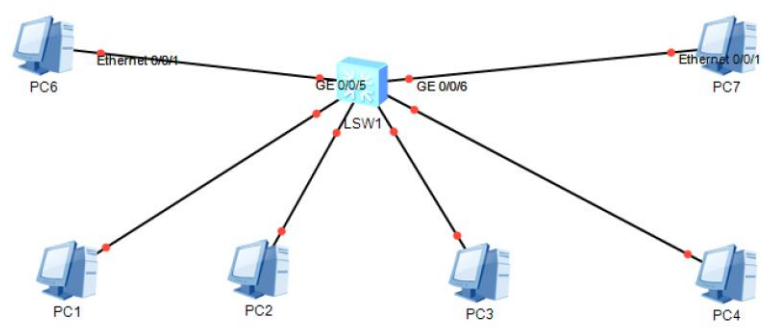
1. Record the MAC addresses of the two laptops in Table 2-6.

Laptop	MAC Adress
PC-BOOK-10-1	5489-9863-5D78
PC-BOOK-30-1	5489-9800-37E2

2. Configure MAC Address-based VLANs

Answer the following questions:

Q2-2.1. Paste the screenshot of the created topology.



Q2-2.2. Paste the table of MAC addresses of the two laptops you recorded.

主机名:

PC-BOOK-10-1

MAC 地址:

54-89-98-63-5D-78

IPv4 配置

☒ 静态

☐ DHCP

☐ 自动获取 DNS 服务器地址

IP 地址:

192 . 168 . 10 . 18

DNS1:

0 . 0 . 0 . 0

子网掩码:

255 . 255 . 255 . 0

DNS2:

0 . 0 . 0 . 0

网关:

0 . 0 . 0 . 0

主机名:

PC-BOOK-30-1

MAC 地址:

54-89-98-00-37-E2

IPv4 配置

☒ 静态

☐ DHCP

☐ 自动获取 DNS 服务器地址

IP 地址:

192 . 168 . 30 . 18

DNS1:

0 . 0 . 0 . 0

子网掩码:

255 . 255 . 255 . 0

DNS2:

0 . 0 . 0 . 0

网关:

0 . 0 . 0 . 0

Q2-2.3. Paste the screenshot of the configuration information of all MAC address-based VLANs.

```
[LSW1]display mac-vlan mac-address all
-----
MAC Address      MASK              VLAN    Priority
-----
5489-9863-5d78   ffff-ffff-ffff   10      0
5489-9800-37e2   ffff-ffff-ffff   30      0

Total MAC VLAN address count: 2

[LSW1]display mac-vlan vlan 10
-----
MAC Address      MASK              VLAN    Priority
-----
5489-9863-5d78   ffff-ffff-ffff   10      0

Total MAC VLAN address count: 1

[LSW1]display mac-vlan vlan 30
-----
MAC Address      MASK              VLAN    Priority
-----
5489-9800-37e2   ffff-ffff-ffff   30      0

Total MAC VLAN address count: 1
```

Q2-2.4. When adding members to VLAN 10 and VLAN 30 based on the MAC address, what is the difference between VLAN 10 and VLAN 30 information before and after adding members? Please paste the screenshot of VLAN 10 information before and after adding members.

```
[LSW1]display vlan 10
-----
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
-----

VID  Type    Ports
-----
10   common  UT:GE0/0/9 (D)    GE0/0/10 (D)     GE0/0/11 (D)     GE0/0/12 (D)

VID  Status  Property      MAC-LRN Statistics Description
-----
10   enable  default      enable  disable  VLAN 0010
```

Before

```
[LSW1]display vlan 10
-----
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
-----

VID  Type    Ports
-----
10   common  UT:GEO/0/5 (D)    GE0/0/6 (D)      GE0/0/9 (D)      GE0/0/10 (D)
                        GE0/0/11 (D)     GE0/0/12 (D)

VID  Status  Property    MAC-LRN  Statistics  Description
-----
10   enable  default    enable   disable    VLAN 0010
```

After

As the graphs shown, GEO0/0/5 and GEO0/0/6 are added to the ports of VLAN10, which indicates that PC-BOOK-10-1 and PC-BOOK-30-1 can reach the VLAN10.

Step 5: Verify the configurations.

Answer the following questions:

Q2-2.5. Can PC-10-1 ping PC-BOOK10-1 each other? Please paste the screenshot of the ping result. Yes ,they can.

```
ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=128 time=63 ms
From 192.168.10.11: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=4 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/43/63 ms
```

PC-BOOK-10-1

```
PC>ping 192.168.10.18

ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time<1 ms
```

PC-10-1

Q2-2.6. Can PC-30-1 ping PC-BOOK30-1 each other? Please paste the screenshot of the ping result.

Yes, they can.

```
PC>ping 192.168.30.18

Ping 192.168.30.18: 32 data bytes, Press Ctrl_C to break
From 192.168.30.18: bytes=32 seq=1 ttl=128 time=32 ms
From 192.168.30.18: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=4 ttl=128 time=62 ms
From 192.168.30.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.18 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/47/62 ms
```

PC-30-1

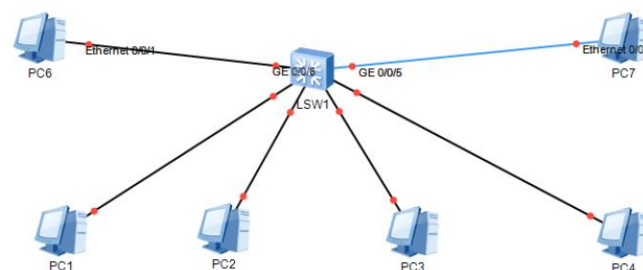
```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=128 time=46 ms
From 192.168.30.11: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.11: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.30.11: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.30.11: bytes=32 seq=5 ttl=128 time=32 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/40/47 ms
```

PC-BOOK-30-1

Q2-2.7. Paste the screenshot of the new topology.



Q2-2.8. Can PC-10-1 ping PC-BOOK10-1 each other? Please paste the screenshot of the ping result.

Yes , they can.

```
PC>ping 192.168.10.18

Ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time<1 ms

--- 192.168.10.18 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 0/0/0 ms
```

PC-10-1

```
PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.10.11: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.10.11: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=5 ttl=128 time=31 ms

--- 192.168.10.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/46/63 ms
```

PC-BOOK-10-1

Q2-2.9. Can PC-30-1 ping PC-BOOK30-1 each other? Please paste the screenshot of the ping result.

Yes, they can.

```
PC>ping 192.168.30.18

Ping 192.168.30.18: 32 data bytes, Press Ctrl_C to break
From 192.168.30.18: bytes=32 seq=1 ttl=128 time=32 ms
From 192.168.30.18: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=4 ttl=128 time=62 ms
From 192.168.30.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.18 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/47/62 ms
```

PC-30-1

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=128 time=46 ms
From 192.168.30.11: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.11: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.30.11: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.30.11: bytes=32 seq=5 ttl=128 time=32 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/40/47 ms
```

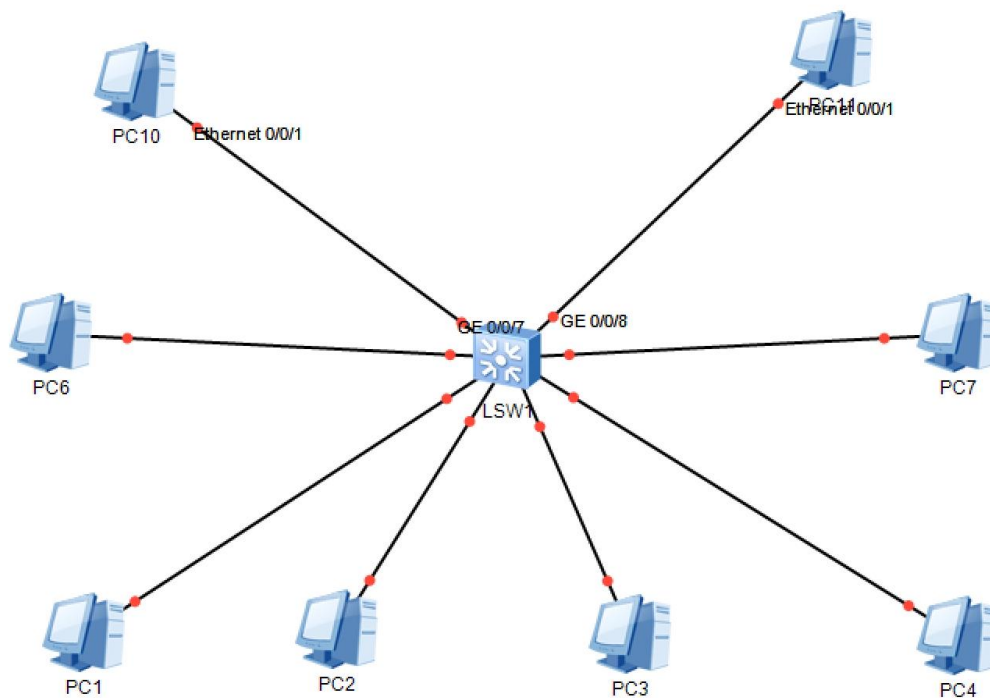
PC-BOOK-30-1

LAB-2-3: IP SUBNET-BASED VLAN CONFIGURATION

Step 4: Configure IP subnet-based VLANs.

Answer the following questions:

Q2-3.1. Paste the screenshot of the created topology.



Q2-3.2. Paste the screenshot of the configuration information of all IP subnet-based VLANs.

```
[LSW1]display ip-subnet-vlan vlan all
-----
Vlan    Index  IPAddress          SubnetMask          Priority
-----
10      1       192.168.10.0       255.255.255.0       1
30      1       192.168.30.0       255.255.255.0       1
-----
ip-subnet-vlan count: 2          total count: 2

[LSW1]display ip-subnet-vlan vlan 10
-----
Vlan    Index  IPAddress          SubnetMask          Priority
-----
10      1       192.168.10.0       255.255.255.0       1
-----
ip-subnet-vlan count: 1          total count: 2
```

```
[LSW1]display ip-subnet-vlan vlan 10
```

Vlan	Index	IpAddress	SubnetMask	Priority
10	1	192.168.10.0	255.255.255.0	1

```
ip-subnet-vlan count: 1          total count: 2
```

```
[LSW1]display ip-subnet-vlan vlan 30
```

Vlan	Index	IpAddress	SubnetMask	Priority
30	1	192.168.30.0	255.255.255.0	1

```
ip-subnet-vlan count: 1          total count: 2
```

```
[LSW1]display vlan
```

The total number of vlans is : 3

```
U: Up;          D: Down;          TG: Tagged;      UT: Untagged;
MP: Vlan-mapping;  ST: Vlan-stacking;
#: ProtocolTransparent-vlan;  *: Management-vlan;
```

VID	Type	Ports
1	common	UT:GE0/0/1(D) GE0/0/2(D) GE0/0/3(D) GE0/0/4(D) GE0/0/5(D) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/17(D) GE0/0/18(D) GE0/0/19(D) GE0/0/20(D) GE0/0/21(D) GE0/0/22(D) GE0/0/23(D) GE0/0/24(D)
10	common	UT:GE0/0/5(D) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/9(D) GE0/0/10(D) GE0/0/11(D) GE0/0/12(D)
30	common	UT:GE0/0/5(D) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/13(D) GE0/0/14(D) GE0/0/15(D) GE0/0/16(D)

VID	Status	Property	MAC-LRN	Statistics	Description
1	enable	default	enable	disable	VLAN 0001
10	enable	default	enable	disable	VLAN 0010
30	enable	default	enable	disable	VLAN 0030

```
[LSW1]display vlan summary
```

static vlan:
Total 3 static vlan.
1 10 30

dynamic vlan:
Total 0 dynamic vlan.

reserved vlan:
Total 0 reserved vlan.

```
[LSW1]display port vlan
```

Port	Link Type	PVID	Trunk VLAN List
------	-----------	------	-----------------

GigabitEthernet0/0/1	hybrid	1	-
GigabitEthernet0/0/2	hybrid	1	-
GigabitEthernet0/0/3	hybrid	1	-
GigabitEthernet0/0/4	hybrid	1	-
GigabitEthernet0/0/5	hybrid	1	-
GigabitEthernet0/0/6	hybrid	1	-
GigabitEthernet0/0/7	hybrid	1	-
GigabitEthernet0/0/8	hybrid	1	-
GigabitEthernet0/0/9	access	10	-
GigabitEthernet0/0/10	access	10	-
GigabitEthernet0/0/11	access	10	-
GigabitEthernet0/0/12	access	10	-
GigabitEthernet0/0/13	access	30	-
GigabitEthernet0/0/14	access	30	-
GigabitEthernet0/0/15	access	30	-
GigabitEthernet0/0/16	access	30	-
GigabitEthernet0/0/17	hybrid	1	-
GigabitEthernet0/0/18	hybrid	1	-
GigabitEthernet0/0/19	hybrid	1	-
GigabitEthernet0/0/20	hybrid	1	-
GigabitEthernet0/0/21	hybrid	1	-
GigabitEthernet0/0/22	hybrid	1	-
GigabitEthernet0/0/23	hybrid	1	-
GigabitEthernet0/0/24	hybrid	1	-

Step 5: Verify the configurations.

Q2-3.3. Can PC-10-1, PC-BOOK10-1 and PC-10-A ping each other? Please paste the screenshot of the ping result.

Yes, they can.

```
PC>ping 192.168.10.18

Ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time=62 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.18 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
    round-trip min/avg/max = 31/43/62 ms

PC>ping 192.168.10.20

Ping 192.168.10.20: 32 data bytes, Press Ctrl_C to break
From 192.168.10.20: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=2 ttl=128 time=32 ms
From 192.168.10.20: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.10.20: bytes=32 seq=4 ttl=128 time=63 ms
From 192.168.10.20: bytes=32 seq=5 ttl=128 time=46 ms

--- 192.168.10.20 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
    round-trip min/avg/max = 32/46/63 ms
```

```
Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=128 time=63 ms
From 192.168.10.11: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=4 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=5 ttl=128 time=47 ms
```

```
--- 192.168.10.11 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 31/43/63 ms
```

```
PC>ping 192.168.10.18
```

```
Ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time<1 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time<1 ms
```

```
--- 192.168.10.18 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 0/0/0 ms
```

```
PC>ping 192.168.10.20
```

```
Ping 192.168.10.20: 32 data bytes, Press Ctrl_C to break
From 192.168.10.20: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=4 ttl=128 time=46 ms
From 192.168.10.20: bytes=32 seq=5 ttl=128 time=63 ms
```

```
--- 192.168.10.20 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 46/50/63 ms
```

```
PC>ping 192.168.10.11
```

```
Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=3 ttl=128 time=62 ms
From 192.168.10.11: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=5 ttl=128 time=46 ms
```

```
--- 192.168.10.11 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 31/46/62 ms
```

```
PC>ping 192.168.10.18
```

```
Ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time=62 ms
```

```
--- 192.168.10.18 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 31/46/62 ms
```

Q2-3.4. Can PC-30-1, PC-BOOK30-1 and PC-30-A ping each other? Please paste the screenshot of the ping result.

Yes ,they can.

```
PC>ping 192.168.30.18

Ping 192.168.30.18: 32 data bytes, Press Ctrl_C to break
From 192.168.30.18: bytes=32 seq=1 ttl=128 time=46 ms
From 192.168.30.18: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.30.18: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.30.18: bytes=32 seq=4 ttl=128 time=63 ms
From 192.168.30.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.18 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 46/53/63 ms

PC>ping 192.168.30.20

Ping 192.168.30.20: 32 data bytes, Press Ctrl_C to break
From 192.168.30.20: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.30.20: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.20 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/44/47 ms
```

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=128 time=32 ms
From 192.168.30.11: bytes=32 seq=2 ttl=128 time=32 ms
From 192.168.30.11: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.30.11: bytes=32 seq=5 ttl=128 time=62 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/40/62 ms

PC>ping 192.168.30.20

Ping 192.168.30.20: 32 data bytes, Press Ctrl_C to break
From 192.168.30.20: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.30.20: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=5 ttl=128 time=46 ms

--- 192.168.30.20 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 46/46/47 ms
```



```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.30.11: bytes=32 seq=2 ttl=128 time=31 ms
From 192.168.30.11: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=128 time=62 ms
From 192.168.30.11: bytes=32 seq=5 ttl=128 time=31 ms

--- 192.168.30.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/43/62 ms

PC>ping 192.168.30.18

Ping 192.168.30.18: 32 data bytes, Press Ctrl_C to break
From 192.168.30.18: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.30.18: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=3 ttl=128 time=63 ms
From 192.168.30.18: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.18 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/47/63 ms
```

Q2-3.5. Can PC-10-1, PC-BOOK10-1 and PC-30-A ping each other? Please paste the screenshot of the ping result. Yes they can.

```
PC>ping 192.168.10.18

Ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time=62 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.18 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/43/62 ms

PC>ping 192.168.10.20

Ping 192.168.10.20: 32 data bytes, Press Ctrl_C to break
From 192.168.10.20: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=2 ttl=128 time=32 ms
From 192.168.10.20: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.10.20: bytes=32 seq=4 ttl=128 time=63 ms
From 192.168.10.20: bytes=32 seq=5 ttl=128 time=46 ms

--- 192.168.10.20 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 32/46/63 ms
```



```

PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.10.11: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.10.11: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.10.11: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=5 ttl=128 time=31 ms

--- 192.168.10.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/46/63 ms

PC>ping 192.168.10.20

Ping 192.168.10.20: 32 data bytes, Press Ctrl_C to break
From 192.168.10.20: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.10.20: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.20: bytes=32 seq=4 ttl=128 time=46 ms
From 192.168.10.20: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.20 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 46/49/62 ms

```

```

PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.11: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.10.11: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 32/44/47 ms

PC>ping 192.168.10.18

Ping 192.168.10.18: 32 data bytes, Press Ctrl_C to break
From 192.168.10.18: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.10.18: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.18 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 47/50/63 ms

```

Q2-3.6. Can PC-30-1, PC-BOOK30-1 and PC-10-A ping each other? Please paste the

screenshot of the ping result.

Yes, they can.

```
PC>ping 192.168.30.18

Ping 192.168.30.18: 32 data bytes, Press Ctrl_C to break
From 192.168.30.18: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=2 ttl=128 time=46 ms
From 192.168.30.18: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.18 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 46/46/47 ms

PC>ping 192.168.30.20

Ping 192.168.30.20: 32 data bytes, Press Ctrl_C to break
From 192.168.30.20: bytes=32 seq=1 ttl=128 time=63 ms
From 192.168.30.20: bytes=32 seq=2 ttl=128 time=62 ms
From 192.168.30.20: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=4 ttl=128 time=31 ms
From 192.168.30.20: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.20 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/50/63 ms
```

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.30.11: bytes=32 seq=2 ttl=128 time=62 ms
From 192.168.30.11: bytes=32 seq=3 ttl=128 time=63 ms
From 192.168.30.11: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.30.11: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/50/63 ms

PC>ping 192.168.30.20

Ping 192.168.30.20: 32 data bytes, Press Ctrl_C to break
From 192.168.30.20: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.30.20: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.30.20: bytes=32 seq=3 ttl=128 time=62 ms
From 192.168.30.20: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.30.20: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.20 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 47/56/63 ms
```

```

PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.30.11: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.30.11: bytes=32 seq=3 ttl=128 time=62 ms
From 192.168.30.11: bytes=32 seq=4 ttl=128 time=63 ms
From 192.168.30.11: bytes=32 seq=5 ttl=128 time=32 ms

--- 192.168.30.11 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 32/53/63 ms

PC>ping 192.168.30.18

Ping 192.168.30.18: 32 data bytes, Press Ctrl_C to break
From 192.168.30.18: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.30.18: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.30.18: bytes=32 seq=3 ttl=128 time=62 ms
From 192.168.30.18: bytes=32 seq=4 ttl=128 time=63 ms
From 192.168.30.18: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.30.18 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 47/56/63 ms

```

Q2-3.7. Paste information about ports or interfaces that belong to each VLAN.

```

[LSW1]display vlan
The total number of vlans is : 3
-----
U: Up;           D: Down;           TG: Tagged;      UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
-----

VID  Type    Ports
-----
1    common  UT:GE0/0/1 (D)    GE0/0/2 (D)      GE0/0/3 (D)      GE0/0/4 (D)
                        GE0/0/5 (D)      GE0/0/6 (D)      GE0/0/7 (D)      GE0/0/8 (D)
                        GE0/0/17 (D)     GE0/0/18 (D)     GE0/0/19 (D)     GE0/0/20 (D)
                        GE0/0/21 (D)     GE0/0/22 (D)     GE0/0/23 (D)     GE0/0/24 (D)
10   common  UT:GE0/0/5 (D)    GE0/0/6 (D)      GE0/0/7 (D)      GE0/0/8 (D)
                        GE0/0/9 (D)      GE0/0/10 (D)     GE0/0/11 (D)     GE0/0/12 (D)
30   common  UT:GE0/0/5 (D)    GE0/0/6 (D)      GE0/0/7 (D)      GE0/0/8 (D)
                        GE0/0/13 (D)     GE0/0/14 (D)     GE0/0/15 (D)     GE0/0/16 (D)

VID  Status  Property      MAC-LRN  Statistics  Description
-----
1    enable  default      enable   disable    VLAN 0001
10   enable  default      enable   disable    VLAN 0010
30   enable  default      enable   disable    VLAN 0030

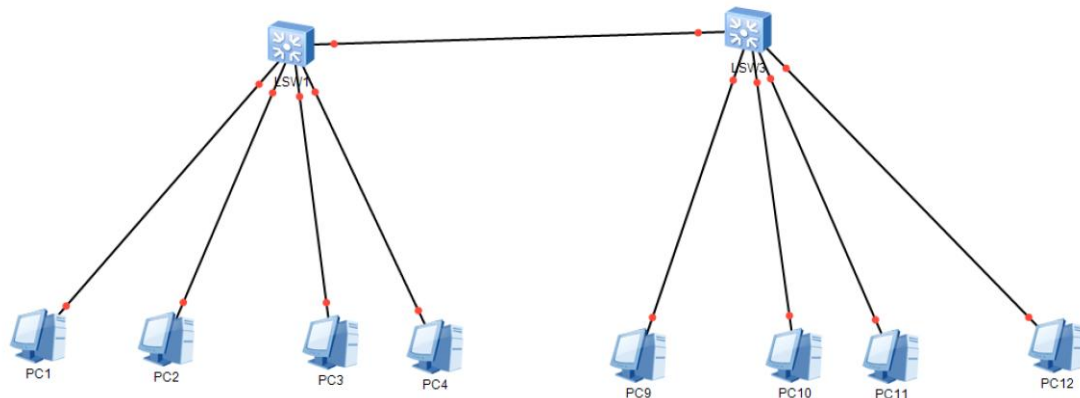
```

LAB-2-4: 802.1Q TRUNK CONFIGURATION

Step 5: Verify the configurations and analyze the traffic on the trunk link.

Answer the following questions:

Q2-4.1. Paste the screenshot of the created topology.



Q2-4.2. Can PC-10-1 and PC-10-3 ping each other? Please paste the screenshot of the ping result.

No, they can't.

```
PC>ping 192.168.10.13

Ping 192.168.10.13: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: Destination host unreachable
From 192.168.10.11: Destination host unreachable
From 192.168.10.11: Destination host unreachable
From 192.168.10.11: Destination host unreachable
From 192.168.10.11: Destination host unreachable

--- 192.168.10.13 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

PC10-1

```
PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.13: Destination host unreachable
From 192.168.10.13: Destination host unreachable
From 192.168.10.13: Destination host unreachable
From 192.168.10.13: Destination host unreachable
From 192.168.10.13: Destination host unreachable

--- 192.168.10.11 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
  100.00% packet loss
```

PC-10-3

Q2-4.3. Can PC-30-1 and PC-30-3 ping each other? Please paste the screenshot of the ping result.

No , they can't.

```
PC>ping 192.168.30.13

Ping 192.168.30.13: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: Destination host unreachable
From 192.168.30.11: Destination host unreachable
From 192.168.30.11: Destination host unreachable
From 192.168.30.11: Destination host unreachable
From 192.168.30.11: Destination host unreachable

--- 192.168.30.13 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
  100.00% packet loss
```

PC30-1

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.13: Destination host unreachable
From 192.168.30.13: Destination host unreachable
From 192.168.30.13: Destination host unreachable
From 192.168.30.13: Destination host unreachable
From 192.168.30.13: Destination host unreachable

--- 192.168.30.11 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
  100.00% packet loss
```

PC30-3

Q2-4.4. Can PC-10-1 and PC-30-3 ping each other? Please paste the screenshot of the ping result.

No ,they can't.

```
PC>ping 192.168.30.13

Ping 192.168.30.13: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: Destination host unreachable
```

PC10-1

```
PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.13: Destination host unreachable
```

PC30-3

Q2-4.5. Are the Ethernet frames captured on interface GE 0/0/24 on switch LSW1 tagged or untagged frames? If they are tagged frames, what are the VLAN IDs? Please paste the screenshot of all field information of two captured Ethernet frames with different VLAN IDs.

The Ethernet frames captured on interface GE 0/0/24 are Tagged frames, because this interface is configured as a Trunk port, and it carries traffic from VLAN 10 and VLAN 30.

As shown in the screenshots below, the 802.1Q VLAN ID fields show different VLANs:

One frame has VLAN ID = 10

Another frame has VLAN ID = 30

```
> Frame 71: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: HuaweiTechno_e9:0c:94 (54:89:98:e9:0c:94), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    Address: Broadcast (ff:ff:ff:ff:ff:ff)
    ....1. .... = LG bit: Locally administered address (this is NOT the factory default)
    ....1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: HuaweiTechno_e9:0c:94 (54:89:98:e9:0c:94)
    Address: HuaweiTechno_e9:0c:94 (54:89:98:e9:0c:94)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: 802.1Q Virtual LAN (0x8100)
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10
  000. .... = Priority: Best Effort (default) (0)
  ...0. .... = DEI: Ineligible
  .... 0000 0000 1010 = ID: 10
  Type: ARP (0x0806)
  Padding: 00000000000000000000000000000000
  Trailer: 00000000
```

```
> Frame 80: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: HuaweiTechno_44:43:3f (54:89:98:44:43:3f), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    Address: Broadcast (ff:ff:ff:ff:ff:ff)
    ....1. .... = LG bit: Locally administered address (this is NOT the factory default)
    ....1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: HuaweiTechno_44:43:3f (54:89:98:44:43:3f)
    Address: HuaweiTechno_44:43:3f (54:89:98:44:43:3f)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: 802.1Q Virtual LAN (0x8100)
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 30
  000. .... = Priority: Best Effort (default) (0)
  ...0. .... = DEI: Ineligible
  .... 0000 0001 1110 = ID: 30
  Type: ARP (0x0806)
  Padding: 00000000000000000000000000000000
  Trailer: 00000000
```

